



Agnel Charities'
Fr. C. Rodrigues Institute of Technology, Vashi, Navi-Mumbai
First Evaluation Mini-Project

IoT-Based Smart System for Fall Detection of Elderly People

Group 01

Vedant Gavhane (3023166)
Parag Dhumal (3023170)
Abhinav Thokale (3023171)
Manisha Kundar (3022130)

Under the Guidance of : Mrs. Akshata Raut
Department of Electronics and Telecommunication Engineering,

Outline

- Problem Statement
- Project Objectives
- System Architecture & Data Flow
- System Components & Hardware Architecture
- Simulation Validation Results
- Literature Survey & Technical Context
- Expected Outcomes & Impact



Problem Statement

Many elderly individuals fall when alone, creating dangerous delays before assistance arrives. Conventional solutions suffer from critical limitations: expensive smartwatches, invasive camera systems that violate privacy, or systems requiring complex installation and maintenance.

Isolation Risk

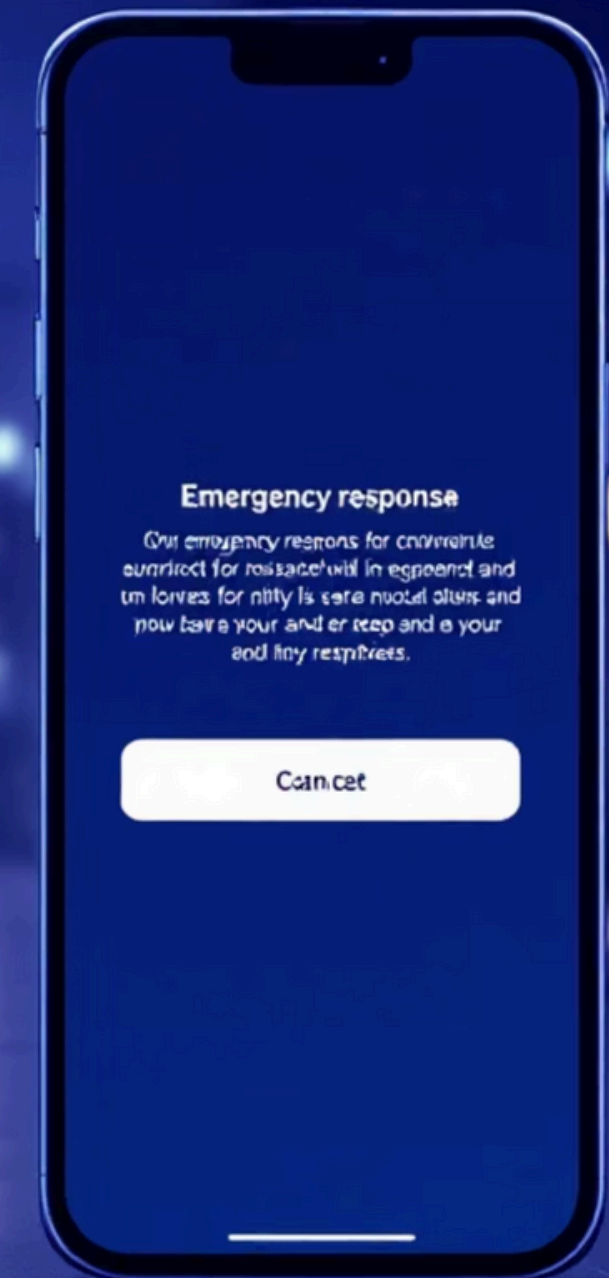
Undetected falls lead to prolonged immobility, muscle breakdown, and psychological trauma.

Cost Barrier

Premium commercial solutions remain inaccessible to most elderly populations and care facilities.

Privacy Concerns

Visual monitoring systems infringe on personal dignity and limit adoption rates significantly.



Project Objectives: Designing a Reliable Wearable System

Falls are a leading cause of injury and mortality among elderly populations. When falls occur in isolation, critical delays in receiving assistance can result in severe complications, permanent disability, or loss of life.

Low-Cost, Wearable Design

Develop a cost-effective, non-intrusive device optimized for continuous monitoring.

Multi-Sensor Integration

Utilize accelerometer and gyroscope (MPU6050) for motion, and MAX30102 for vital signs (HR/SpO₂).

Emergency Communication

Implement GSM (SIM800L) for SMS alerts and GPS (NEO-6M) for precise location sharing.

Efficiency and Privacy

Ensure device portability, energy efficiency (3.7V Li-Po), and local processing for enhanced user privacy.



System Components & Hardware Architecture

Our integrated wearable platform combines multiple sensors for comprehensive fall detection and health monitoring, all coordinated through a central microcontroller with global connectivity.



ESP32-S3 Round Display

1.46-inch processor and display module with integrated WiFi and Bluetooth.



NEO-6M GPS Module

Precision location tracking for emergency responder guidance.



SIM800L GSM Module

Global connectivity for SMS alerts independent of WiFi availability.



MAX30102 Sensor

Simultaneous heart rate and oxygen saturation monitoring.

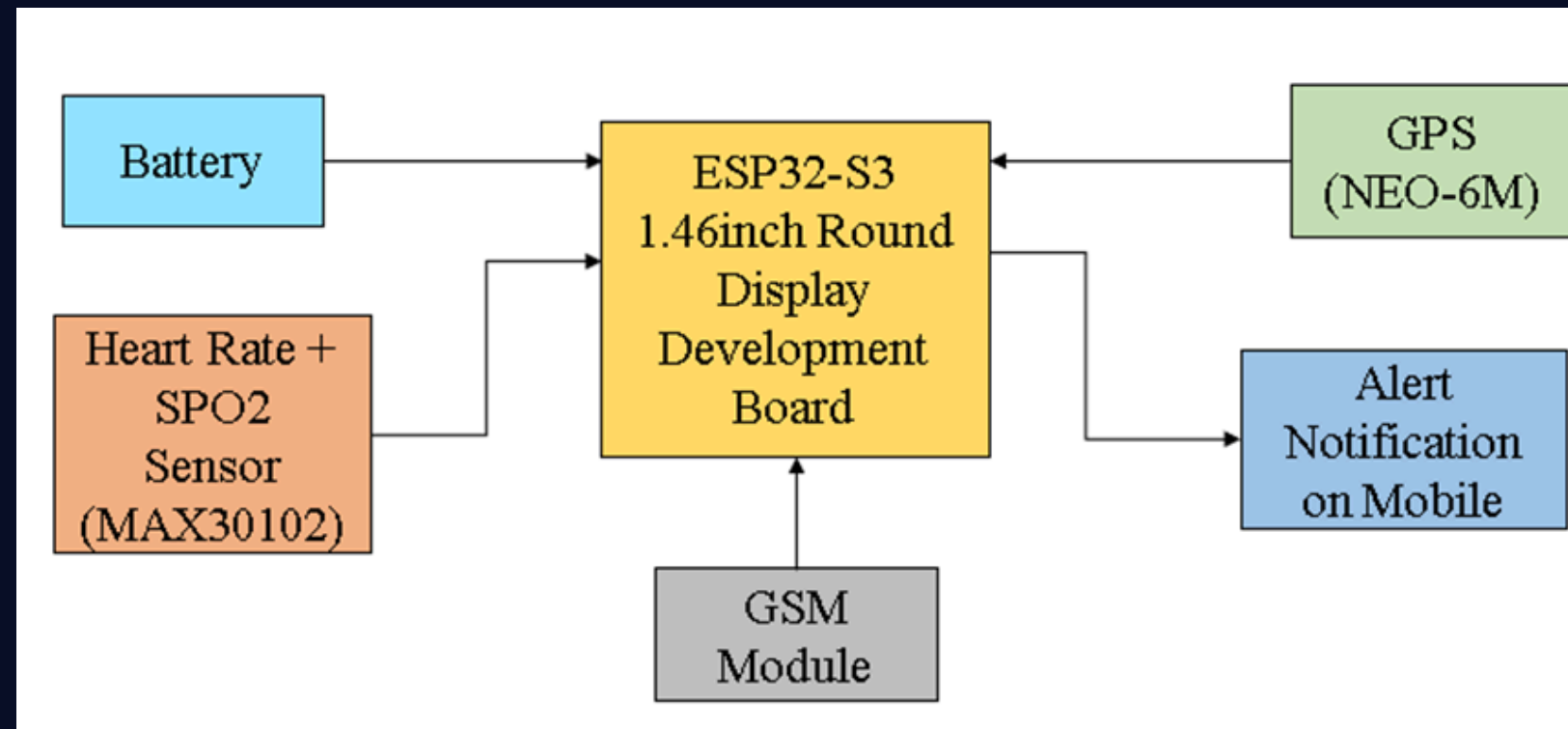


3.7V Lithium Battery

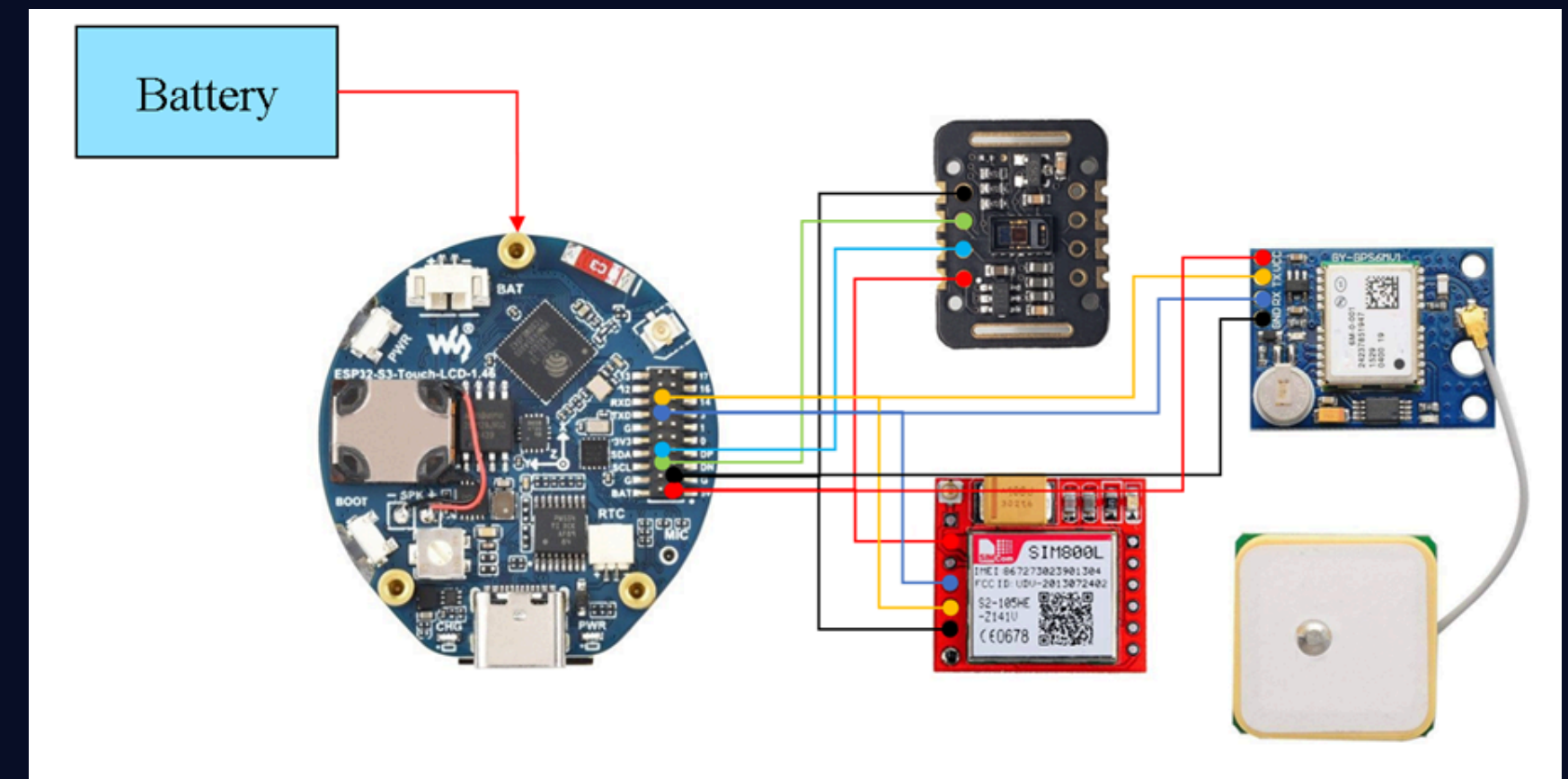
Rechargeable power source ensuring extended daily operation.

System Architecture & Data Flow

The system integrates multiple sensor inputs through the ESP32 microcontroller, processes fall detection algorithms, and transmits alerts via GSM to emergency contacts with precise location data.



Block Diagram of Smart Fall Detection System



Circuit Diagram for Smart Fall Detection System

Simulation Validation Results

Wokwi simulation testing demonstrates the system's ability to reliably distinguish normal motion patterns from fall events through real-time acceleration analysis.

Normal State Operation

MPU6050 accelerometer readings stabilize at approximately 19.19 m/s² during regular movement and static positioning. No alerts triggered.



Gyro & Accelerometer Normal State Simulation

Fall Detection Triggered

Acceleration spike exceeds 26 m/s² threshold when simulating rapid vertical descent. Red alert LED activates immediately and system initiates notification sequence.

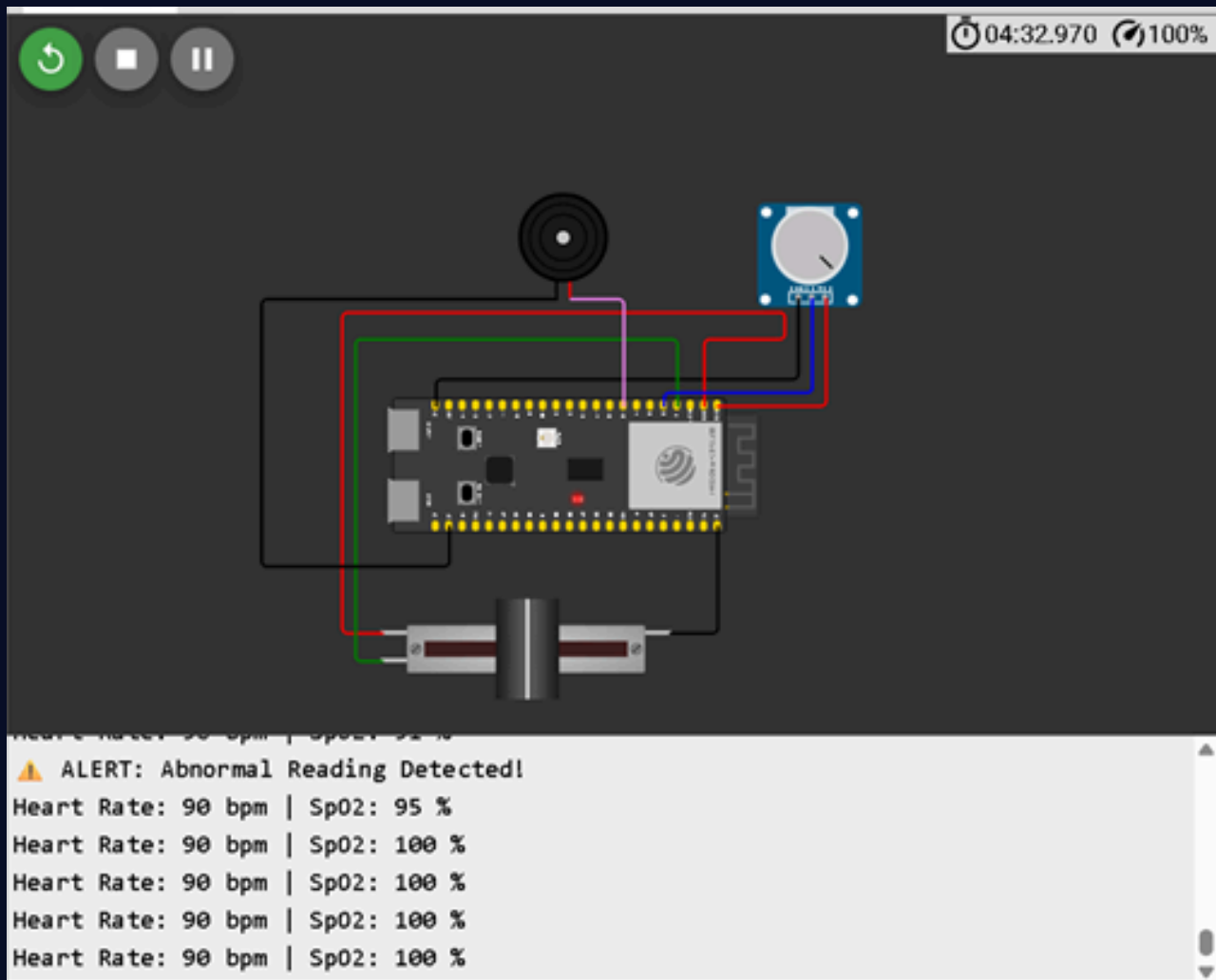


Gyro & Accelerometer Fall Detected Simulation

Simulation Validation Results

Heart Rate & SpO2 are within the Threshold

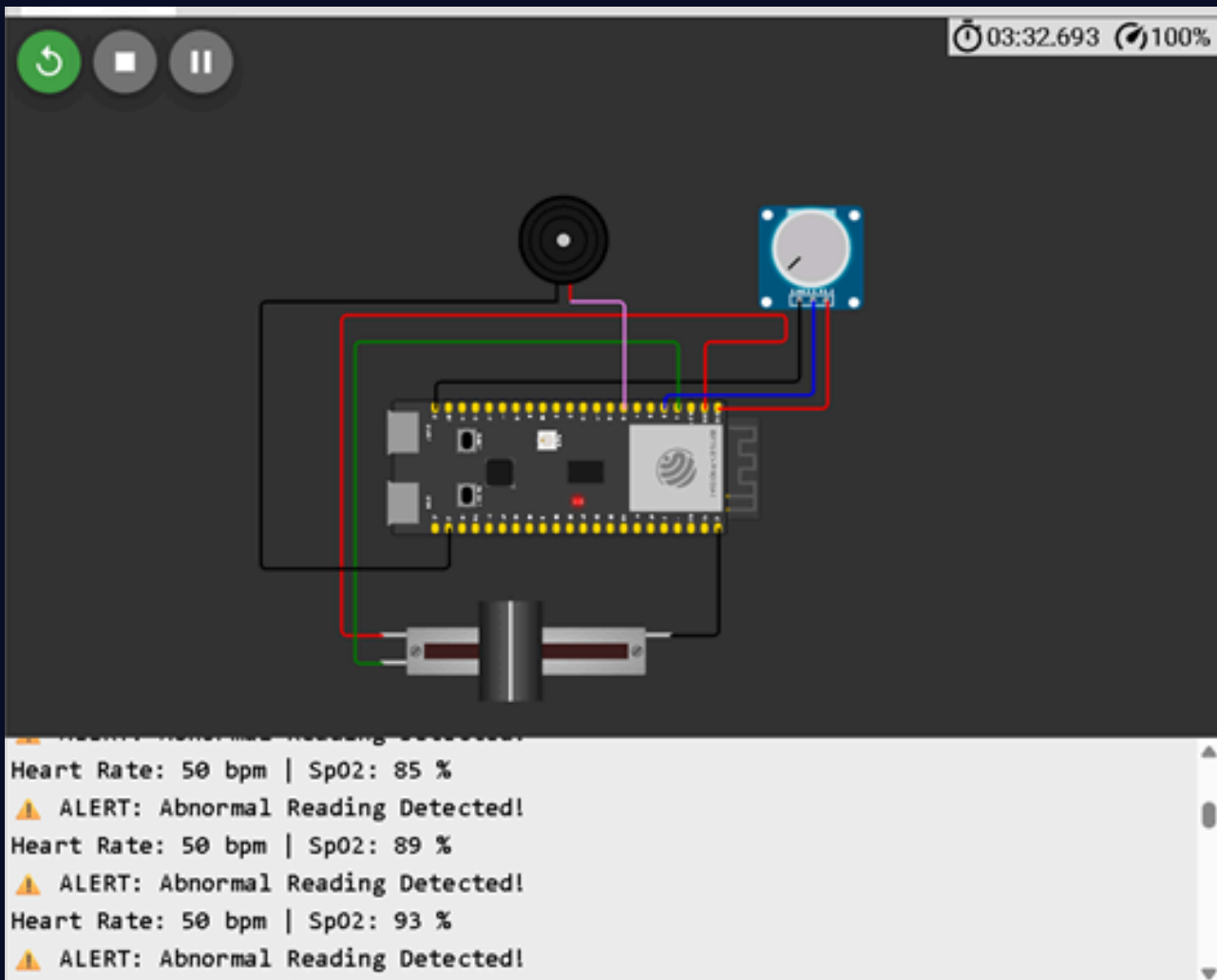
If the heart rate is at 90 bpm and SpO₂ improves to 95–100%, which are within the normal range, so no alert is generated.



Heart Rate & SpO2 is in given Threshold

Heart Rate & SpO2 crosses the Threshold

If the heart rate or SpO₂ value falls outside the normal range, an alert message “⚠️ ALERT: Abnormal Reading Detected!” is displayed, and the buzzer is activated to provide an audible warning.



Heart Rate & SpO2 Crosses the Threshold

Literature Survey & Technical Context

Recent research validates the effectiveness of sensor-based fall detection with machine learning integration, though challenges remain in real-world scalability and cross-population accuracy.

Authors & Year	Methodology	Key Results	Limitations
Awantika Krishna et al. (2024)	MPU6050, ESP32, AI algorithms, IoT (Blynk)	95% sensitivity; reduced false alarms; real-time alerts	Cannot detect seated falls; limited scalability
Shiv Narain Gupta et al. (2024)	Wearable wristband accelerometer & gyroscope; IoT + Android App	Real-time alerts; non-intrusive; ~83% accuracy	Threshold-dependent; population-specific variations
Survey Paper (2024)	ML + IoT comprehensive review	CNN and SVM integration best practices identified	High computational overhead; limited large-scale validation

Expected Outcomes & Impact

This project delivers a practical, scalable fall detection platform that bridges the gap between commercial solutions and actual elderly care needs through affordable hardware, proven algorithms, and immediate emergency response capabilities.

User Autonomy

Non-intrusive wearable respects privacy while enabling independent living with safety assurance.

Cost Efficiency

Affordable component-based design enables deployment across diverse socioeconomic populations.

Scalability

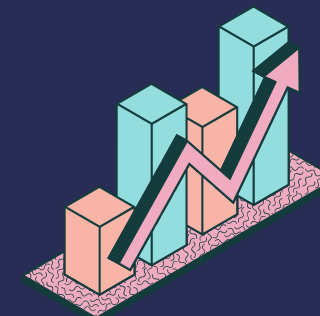
Modular architecture facilitates integration with existing IoT ecosystems and care facility infrastructure.

Safety Enhancement

Rapid detection and immediate caregiver notification minimizes injury severity and recovery time.

Health Integration

Biometric monitoring adds preventive health insights beyond fall detection alone.



GANTT CHART

Task\Months	August	September	October	November
Information / Components collection	✓			
System Design & Planning	✓	✓		
Component Procurement		✓	✓	
Documentation & Report Writing			✓	✓
Sensor Testing		✓		✓

REFERENCES

- [1] Gupta, S. N., Hussain, A., Kumar, A., & Pramanik, I. (2024). IoT based Smart Fall Detection Device for Elderly People. 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT), 1–5. <https://doi.org/10.1109/iceect61758.2024.10739038>
- [2] Krishna, A., Yadav, B., Prakash, D., Dwivedi, R. K., & IEEE. (2024). Designing A Fall Detection System for Elderly People in Smart Healthcare IoT Using Multi Sensor Data Fusion. 2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA). <https://doi.org/10.1109/ICAIQSA64000.2024.10882340>
- [3] J. Akilandeswari and A. Velusamy, “Recent Advances in Fall Detection and Prevention Using Machine Learning and IoT: A Comprehensive Review,” in Proc. Int. Conf. on Emerging Technologies in Engineering Applications (ICETEA), Salem, India, 2025, pp. 1–7. doi: 10.1109/ICETEA64585.2025.11099926

THANK YOU!!!